# System Models

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# System Model

- A model is a simplified reflection of reality to understand, define, quantify, visualize, or simulate empirical objects, phenomena, and physical processes by referencing it
  - Models should represent them in a logical and objective way
  - Modelling is the process of generating a model as a conceptual representation of some object, phenomenon, or physical process

- System models are an important bridge between the requirements analysis and system design processes

# Analysis Model vs. Design Model

## ❒ Analysis model

- The analysis process focuses on aspects of the problem that are visible to users or customers
- The level of abstraction of an analysis model is relatively high

## ❒ Design model

- The design process refines the analysis model by providing detail that will enable the model to be implemented, and creates a new set of model elements that need not be visible to users and customers (visible only to developers)
- The level of abstraction of a design model is relatively low

# System Modelling Perspectives

- Different models present the system from different perspectives
  - External perspective shows the system's context or environment
  - Structural perspective shows the system or data architecture
  - Behavioural perspective shows the behaviour of the system

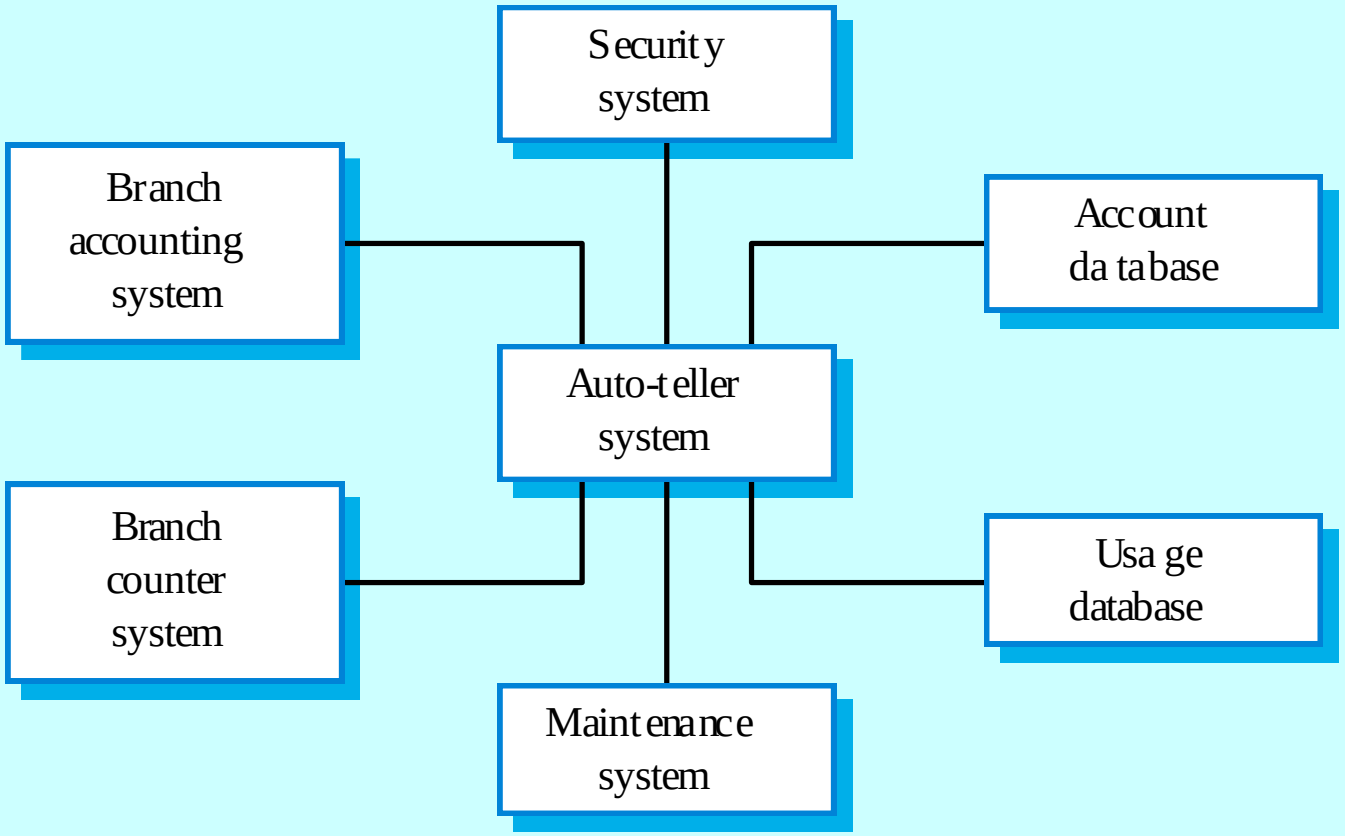
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# Context Models

- At an early stage in the requirements elicitation and analysis process, you should decide on the boundaries of the system
  - This involves working with system stakeholders to distinguish what is the system and what is the system's environment
- In some cases, the boundary may be relatively clear
- In other cases, there is more flexibility, and you decide what constitutes the boundary during the requirements engineering process

# The Context of an ATM System

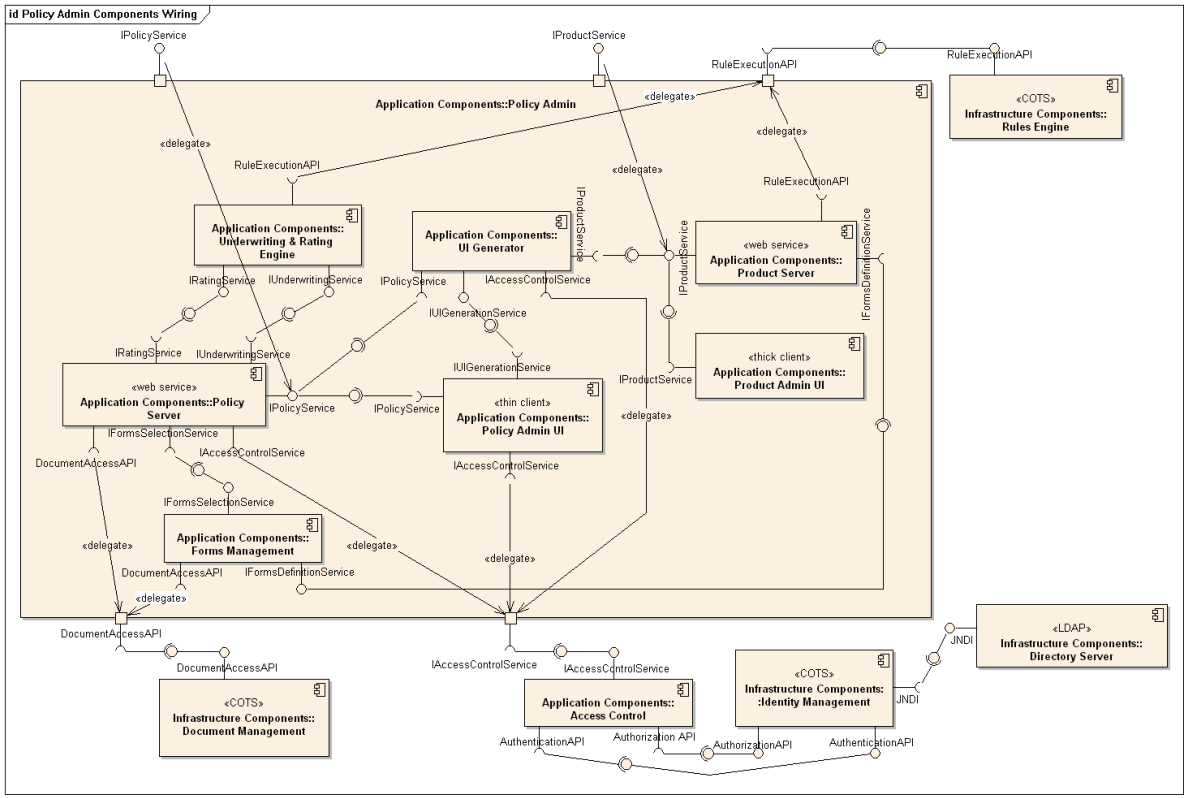


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# Component Diagrams

- A component diagram depicts what components exist inside the system and how they are wired together

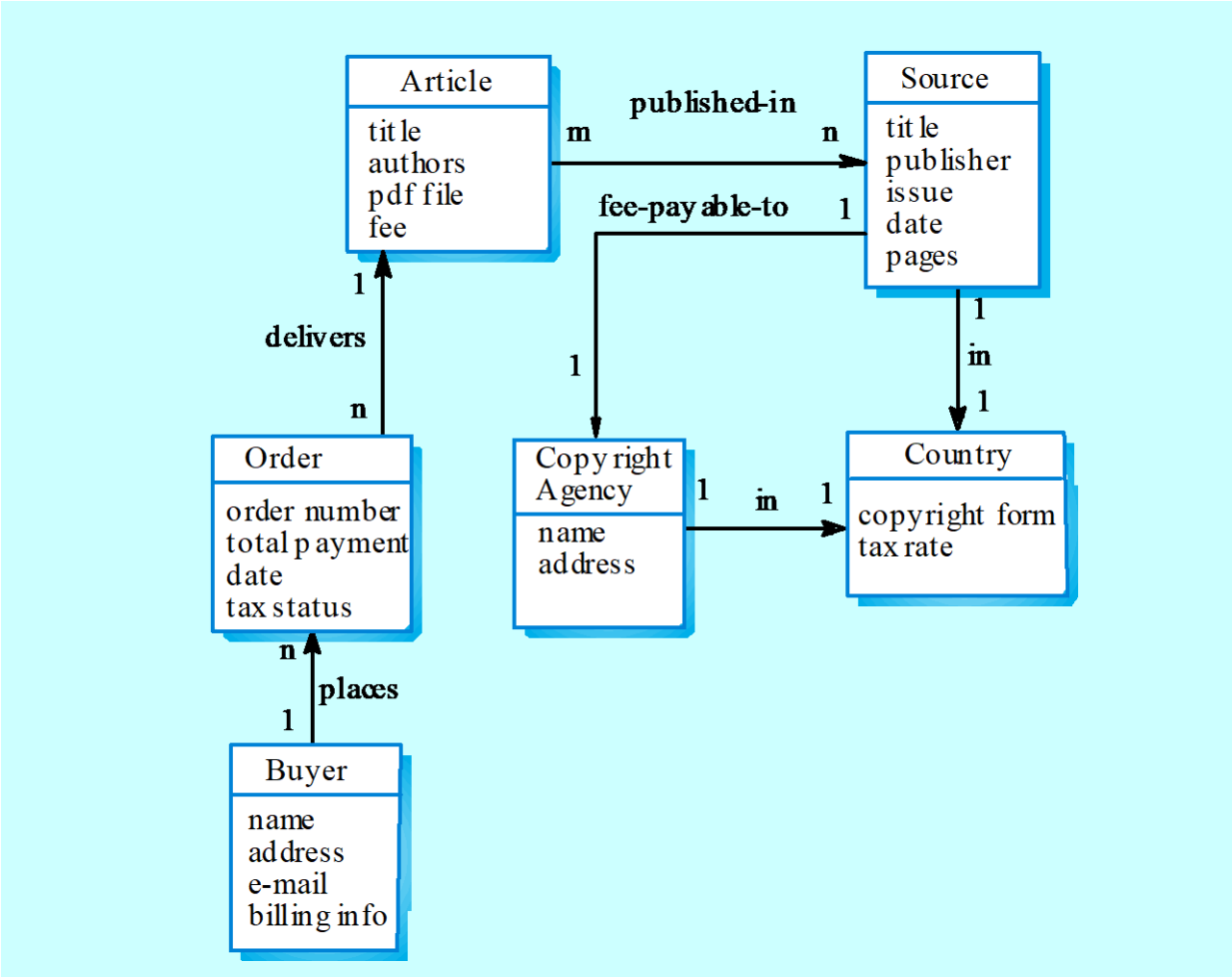


# Data Models

- A data model defines the logical structure of the data processed by the system
  - Sometimes called semantic data model

- An entity-relation-attribute (ERA) modelling is the most widely used technique
  - ERA modelling shows the data entities, their associated attributes, and the relations between these entities

# Library Data Model



# Data Dictionaries

- Like all graphical models, data models lack detail, and you should maintain more detailed descriptions of the entities, relationships, and attributes that are included in the model
  - You can collect these more detailed descriptions in a repository or data dictionary

- A data dictionary is an alphabetical list of the names used in the system models
  - Descriptions of the entities, relationships and attributes are also included

# Data Dictionary Entries

| Name | Description | Type | Date |
|---|---|---|---|
| Article | Details of the published article that may be ordered by people using LIBSYS. | Entity | 30.12.2002 |
| authors | The names of the authors of the article who may be due a share of the fee. | Attribute | 30.12.2002 |
| Buyer | The person or organisation that orders a copy of the article. | Entity | 30.12.2002 |
| fee-payable-to | A 1:1 relationship between Article and the Copyright Agency who should be paid the copyright fee. | Relation | 29.12.2002 |
| Address (Buyer) | The address of the buyer. This is used to any paper billing information that is required. | Attribute | 31.12.2002 |

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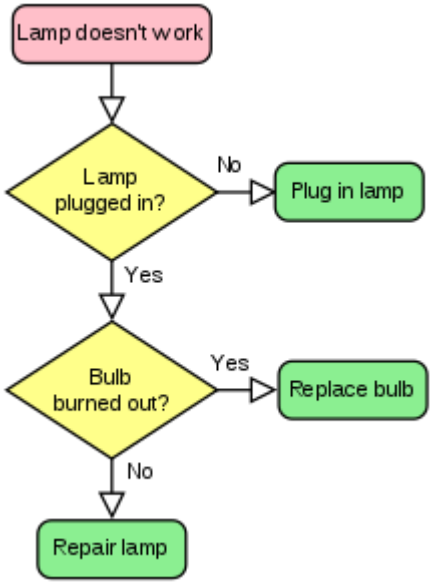
# Behavioural Models

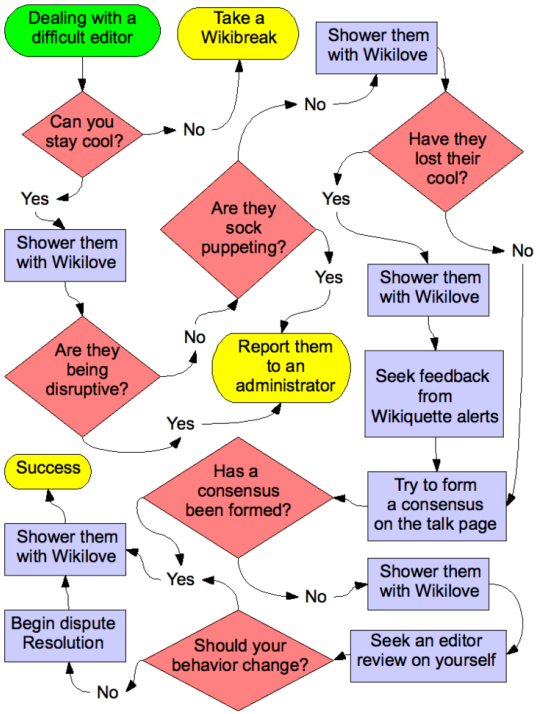
- Behavioural models are used to describe the overall behaviour of a system

- A widely used behavioural models are
  - Flow chart that shows a workflow or process
  - Data-flow model that shows how data is processed by a system
  - State machine model that shows the system's response to events

# Flow Charts

- Flow charts visualize what is going on and thereby help understand a process, and perhaps also find less-obvious features within the process

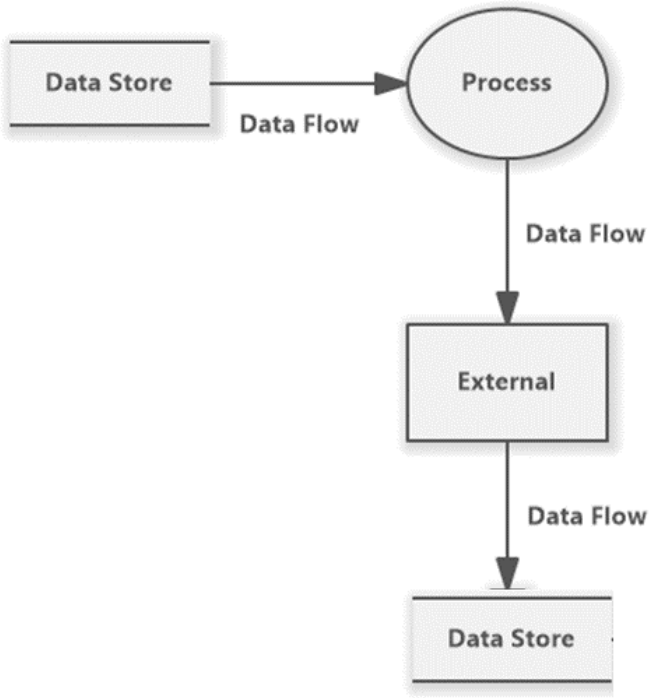


# Data-Flow Models

- Data-flow models are an intuitive way of showing how data is processed by a system
  - They are used to show how data flows through a sequence of processing steps

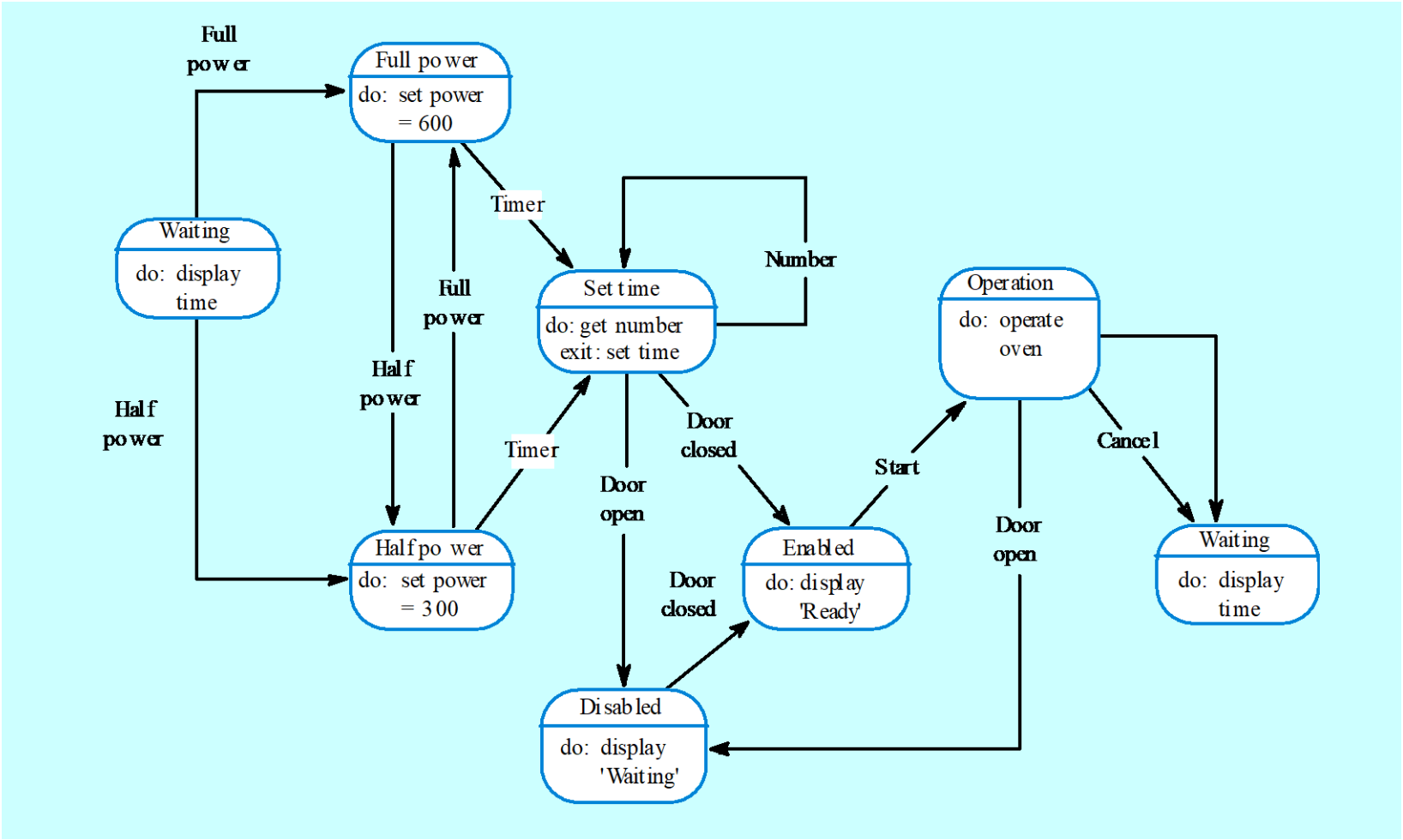
- Data-flow diagrams (DFDs)
  - Data stores: rectangles
  - Data movement: labelled arrows
  - Functional processing
    - Also known as transformation
    - Circles or rounded rectangles



# State Machine Models

- A state machine model describes how a system responds to internal or external events
  - They show system states and events that cause transitions from one state to another
  - They do not show the flow of data within the system
  - They are often used for modelling real-time systems because these systems are often driven by stimuli from the system's environment

# Microwave Oven Model



# Problems with State Machine Models

- State machine models are relatively complex and difficult to understand when compared to other models

- The number of possible states increases rapidly
  - For large system models, one way to manage a large number of states is by using the notion of a superstate that encapsulates a number of separate states
  - This superstate looks like a single state on a high-level model but is then expanded in more detail on a separate diagram

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# Object Models

- Object models may be used to represent both system data and its processing
  - In this respect, they combine some of the uses of data-flow model and semantic data models
- Object models are natural ways of reflecting the real-world entities that are manipulated by the system
  - This is particularly true when the system possesses information about tangible entities, such as cars, aircraft, or books
- More abstract, higher-level entities, are harder to model as objects
  - E.g., a library, a medical record system, or a word processor

# Object Models

- Developing object models during requirements analysis usually simplifies the transition to object-oriented design and programming

- However, end-users of a system often find object models unnatural and difficult to understand
  - They may prefer to adopt a more functional, data-processing view
  - Therefore, it is sometimes helpful to supplement object models with data-flow models that show the end-to-end data processing in the system

# Objects and Classes

- A class is an abstraction over a set of entities that identifies common attributes and the services or operations that are provided by each object
  - Objects are instantiations of classes

- Generally, the object models focus on classes and their relationships


Q & A

# 모델링에 대하여 ...

- 모델은 머리 속에 존재하고, 코드는 컴퓨터 속에 존재한다
  - **(분석)** 모델은 문제에 대한 통찰이다
  - **(설계)** 모델은 해법에 대한 전략이다
- 모델은 객관화되고 표출되어야 한다
  - 설명될 수 없다면 모르는 것이거나 불완전한 것이다
- 같은 문제라 하더라도 모델은 사람마다 다르다
  - 관점에 따라 문제에 대한 통찰**(understanding)**과 해법에 대한 전략**(strategy)**가 다를 수 있기 때문이다
- 그러나 공통적인 원칙은 존재한다
  - 추상화**(abstraction)**, 모듈화**(modularity)**, 명확성**(clarity)**, 이해성**(understandability)**, **…**
- 일반적으로 구조적 모델링을 먼저 수행하고, 그 다음에 행위적 모델링을 수행한다